Market Reactions to FOMC Announcements

Master of Quantitative Economics

University of California, Los Angeles

Kojo Manu

Chris Surro, Advisor

03/11/2026

Table of Contents

1. Abstract 3
2. Introduction 4
3. Literature Review 5
4. Data 7
5. Results 8
6. Conclusion 17
7. References 19

1. Abstract

This paper examines how financial markets respond to Federal Open Market Committee (FOMC) communication by combining high-frequency monetary policy surprises from the U.S. Monetary Policy Event-Study Database (USMPD) with FinBERT-based sentiment scores derived from post-meeting statements. I estimate event-study regressions of S&P 500 returns and Treasury yields on monetary event (ME) surprises and statement sentiment to assess whether linguistic tone contains incremental information beyond measurable policy shocks.

ME surprises generate the expected cross-asset responses: contractionary shocks significantly lower equity prices and raise Treasury yields across maturities. Importantly, sentiment provides incremental explanatory power for S&P 500 returns even after controlling for policy surprises, with more optimistic communication associated with higher equity returns. In contrast, sentiment does not independently affect 10-year Treasury yields. Instead, statement tone conditions yield responses to directional policy shocks. Optimistic communication amplifies the yield response to hawkish surprises, while pessimistic tone attenuates it. No comparable interaction effect appears in equities, where tone operates as a direct and independent driver.

2. Introduction

Financial markets respond within minutes to new information about the path of monetary policy. Among the most closely watched sources of such information are Federal Open Market Committee (FOMC) post meeting statements and the Chair's subsequent press conferences. These communications shape expectations about the future path of interest rates, inflation, and economic activity, and therefore influence asset prices across the yield curve and in equity markets. Understanding how markets interpret the content of FOMC communication is central to both monetary and financial economics.

While a large literature documents that monetary policy "surprises" move asset prices, recent work emphasizes that central bank communication conveys more than just mechanical changes in policy rates. For the purposes of this paper, monetary policy surprise is defined as the unexpected change in short-term interest rate expectations revealed during FOMC communication events, measured using high-frequency movements in money market futures around those events. The tone and language of FOMC statements may shift expectations about the economic outlook or the reaction function of the Federal Reserve, even when the policy rate itself is unchanged. This raises two related questions: (1) how do financial markets react to the sentiment embedded in FOMC statements, and (2) how does that reaction differ from, or interact with, measured monetary policy surprises?

This paper combines two novel data sources to address these questions. First, I use sentiment measures constructed from FOMC post-meeting statements, following *Predicting the Reaction of Financial Markets to Federal Open Market Committee Post-Meeting Statements* (Osowska & Wójcik, 2023). Their dataset quantifies the tone of post-meeting statements using textual analysis techniques. However, as their dataset only goes through June 2023, I extend it to the current most recent statement by applying their methodology, specifically the FinBERT model developed by Yang et al. (2020), to all subsequent FOMC statements, yielding a complete and current sentiment series. Second, I incorporate high-frequency monetary policy surprises from *Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database* (Acosta et al., 2025), which decomposes communication into distinct components: statements, press conferences, and the combined monetary policy event, and then measures the unexpected changes in interest rate expectations around each event.

A central question in monetary policy research is whether how the Fed communicates matters independently of what it decides. Standard high-frequency surprise measures capture the market's reassessment of the rate path around FOMC events, but they do not distinguish between the mechanical effect of the policy decision itself, and the forward guidance embedded in the accompanying statement. If sentiment in FOMC communications carries information beyond what is reflected in rate surprises (i.e. shifting yield curves or equity prices in ways that surprise measures do not capture) then the tone of Fed communication is itself a policy tool, not merely a description of policy. Merging these datasets allows me to disentangle the effects of textual sentiment from market-based measures of policy surprises. I estimate high-frequency event-study regressions of changes in Treasury yields across maturities and the S&P 500 index on both sentiment and surprise measures. In addition, I examine movements in Eurodollar and SOFR futures to classify whether markets interpret communication as hawkish or dovish, and to assess whether sentiment independently shifts rate expectations beyond what is captured by standard surprise measures.

This approach contributes to the literature in two ways. First, it evaluates whether textual sentiment contains incremental information for asset prices after controlling for conventional monetary policy surprises. Second, by comparing responses across Treasury maturities and equity markets, it sheds light on whether sentiment primarily operates through expected policy rates, risk premia, or broader economic outlook channels.

The results provide evidence on how financial markets process central bank communication in real time and whether the tone of FOMC statements systematically shapes expectations beyond measurable policy shocks. These findings have implications for the design and transparency of monetary policy communication, as well as for investors seeking to understand and anticipate market reactions to FOMC events.

3. Literature Review

The study of financial market reactions to Federal Open Market Committee (FOMC) announcements has undergone a significant transformation, shifting from a narrow focus on interest rate decisions to a broader analysis of communication strategies, forward guidance, and linguistic tone. A central challenge throughout this literature has been isolating the truly unexpected component of monetary policy, separating the Fed's systematic reaction to economic conditions from exogenous policy shifts. Romer and Romer (2004) addressed this by using the

Fed's own internal Greenbook forecasts to purge rate changes of their predictable component, demonstrating that truly exogenous tightening has substantially larger effects on output and inflation than naive measures suggest. While their approach leverages the Fed's information set, it is less suited to studying high-frequency asset price reactions within narrow event windows. Kuttner (2001) offered a market-based solution, isolating policy surprises using federal funds futures and showing that only the unanticipated component of rate changes significantly moves Treasury yields. This high-frequency approach, refined and extended by Acosta et al. (2025) into a multidimensional surprise framework, forms the methodological foundation of this paper.

As the Fed increasingly relied on forward guidance and communication as policy tools, particularly during the zero lower bound period following the 2008 financial crisis, researchers began to question whether rate surprises alone were sufficient to capture the full informational content of FOMC events. Gürkaynak et al. (2004) provided early evidence that FOMC announcements contain at least two distinct factors: a surprise in the current target rate and a "path factor" capturing revisions to the expected future rate path. Their work showed that the path factor, not the target surprise, drives the bulk of Treasury yield movements at longer maturities, motivating the multidimensional surprise measures used in contemporary work. More recently, Acosta et al. (2025) formalized this decomposition and documented that the press conference has surpassed the written statement as the primary driver of policy news and market volatility, emphasizing the use of a "monetary event" (ME) window that aggregates news from both communication channels.

A separate but related strand of the literature asks whether the linguistic content of Fed communication carries information beyond quantifiable rate surprises. The evidence here is more mixed. Lucca and Trebbi (2009) constructed measures of FOMC communication hawkishness using computational linguistics and found that changes in the directional tone of statements predict subsequent short-term interest rate movements, suggesting that textual content contains forward-looking policy information. However, dictionary-based approaches, such as those using the Loughran-McDonald financial sentiment lexicon, face a fundamental identification challenge: disentangling the independent effect of tone from the information already embedded in the policy action itself is difficult when both are released simultaneously, making it hard to draw clean conclusions about whether tone carries incremental information beyond the rate surprise.

The advent of large pre-trained language models has sharpened this analysis considerably. Yang et al. (2020) developed FinBERT, a finance-specific BERT model trained on corporate reports,

analyst reports, and earnings call transcripts, which captures financial language more accurately than general-purpose or dictionary-based approaches. Applying FinBERT to FOMC statements, Osowska and Wójcik (2023) found that sentiment scores derived from the written statement are among the first or second most important predictors of financial market reactions in the first minute following an announcement. However, the predictive power of these communication-based models is sensitive to economic regimes, with accuracy markedly lower during the COVID-19 pandemic due to extreme volatility and heightened macroeconomic uncertainty.

The transmission of FOMC communication to Treasury markets reveals complex dynamics between nominal yields, real yields, and inflation expectations. While short-term rates respond to immediate policy changes, longer-dated Treasury yields react primarily to revisions in the expected path of policy rates. Recent evidence indicates that hawkish monetary policy surprises have a pronounced negative effect on break-even inflation, particularly at intermediate maturities, suggesting that FOMC announcements signal changes in the Fed's perceived reaction function and long-run inflation targets beyond simple discount rate effects (Acosta et al., 2025). Risk assets, specifically the S&P 500, demonstrate strong negative sensitivity to contractionary surprises through three primary channels: increased risk-free discount rates, reduced investor risk appetite, and lowered dividend expectations (Bernanke & Kuttner, 2005; Acosta et al., 2025).

This paper incorporates high-frequency surprise measurement and textual sentiment analysis, using their combination to further the discussion of whether linguistic tone contains incremental information beyond rate surprises. By jointly estimating the effects of ME surprises and FinBERT sentiment on both equity and Treasury markets, and introducing an interaction between directional policy shocks and statement tone, the results offer evidence that the two channels are complementary rather than redundant, and that their interaction differs systematically across asset classes in ways that prior work has not fully characterized.

4. Data

The core of the monetary policy data is sourced from the U.S. Monetary Policy Event-Study Database (USMPD), a novel dataset providing high-frequency asset price changes around FOMC communication events. This database includes intraday changes for money market futures, Treasury and TIPS yields, stock indices, and exchange rates. The USMPD separates policy news into four distinct event windows: the statement and minutes windows (30 minutes), the press

conference window (70 minutes), and a comprehensive monetary event window that begins 10 minutes before the statement release and 60 minutes after the start of the press conference. When there is no press conference, the statement and monetary event windows are identical.

To quantify unexpected shifts in monetary policy, this study utilizes the multidimensional surprise methodology established by Acosta et al. (2025). This approach identifies the "path factor" of policy news by calculating the first principal component of intraday changes across five money market instruments: the near-term federal funds surprises for the current and next meetings (MP1 and MP2) and three quarterly futures contracts (ED2, ED3, and ED4).

To ensure economic interpretability, the resulting principal components are normalized so that a one unit move corresponds to a one percentage-point effect on the daily change in the one-year Treasury yield. Due to the cessation of LIBOR, the calculation of these surprises transitioned from Eurodollar futures to Secured Overnight Financing Rate (SOFR) futures starting in January 2022 to maintain a consistent measure of long-term policy expectations. Alternatively, some models in the literature, such as those used by Osowska and Wójcik (2023), utilize the Kuttner (2001) approach, which isolates surprises solely from 30-day Federal Funds Futures by rescaling price changes based on the number of days remaining in the month.

The linguistic "tone" of FOMC communication is quantified using the FinBERT pre-trained NLP model, as detailed by Osowska and Wójcik (2023). FinBERT is a finance-specific language model trained on corporate reports, analyst reports, and earnings call transcripts, making it uniquely suited for the Fed's specialized vocabulary and financial jargon. The calculation process involves rigorous text preprocessing of the FOMC statements, including:

- Cleaning: Converting text to a lowercase corpus and removing punctuation, numbers, and standard English stop words.
- Lemmatization and Tokenization: Reducing words to their base forms and separating them into individual units (tokens) for analysis.
- Scoring: The processed text is fed into the FinBERT model, which assigns a sentiment score ranging from -1 (negative/pessimistic) to 1 (positive/optimistic).

5. Results

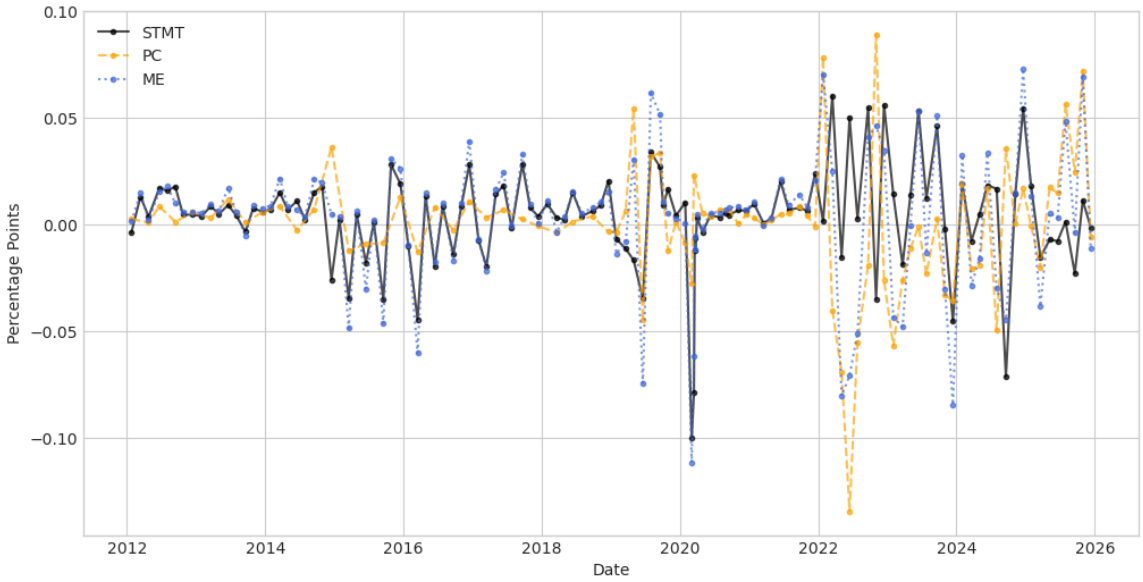
Using the USMPD, I first create Figure 1 following Acosta et al. 2025, which plots the monetary policy surprises derived from FOMC statements (STMT), press conferences (PC), and

broader monetary events (ME) from 2012 to 2026. From 2012 through roughly 2019, surprises across all three communication channels remain relatively small in magnitude and tightly clustered around zero, suggesting that policy decisions were largely anticipated by markets during this period. Occasional deviations occur, particularly in 2015–2016, but these remain moderate and short-lived.

In 2020, there is a pronounced spike in volatility, coinciding with the COVID-19 global pandemic. All three measures exhibit large negative surprises, followed by sharp reversals, reflecting substantial and unexpected policy interventions. This period marks a structural break relative to the preceding years. From 2022 onward, the magnitude and frequency of surprises increase again, with PC having a significant negative surprise in 2022.

STMT and PC surprises exhibit a modest negative correlation (–0.125), indicating a slight inverse relationship between the two communication channels. As shown in Figure 1, this pattern becomes more apparent during periods of heightened volatility, particularly from 2022 onward. Because the FOMC statement is released approximately 30 minutes before the press conference, markets initially respond to the information contained in the written statement. The subsequent press conference often serves to clarify, refine, or in some cases reinterpret that initial signal, as market participants incorporate the Chair's wording, tone, and responses during the Q&A session to update expectations about the future path of monetary policy.

Figure 1 Monetary Policy Surprises by Communication Channel (2012–2026)



The descriptive evidence in Figure 1 highlights substantial time variation in the magnitude and volatility of monetary policy surprises, particularly during periods of economic stress. However, understanding the economic significance of these shocks requires examining their transmission to financial markets. To quantify this relationship, I estimate regressions of S&P 500 returns and Treasury yields on ME surprises within the event window, allowing for a direct assessment of how unexpected policy shifts affect markets.

I first estimate a regression of S&P 500 returns on the ME surprise. As shown in Table 1, the coefficient on ME is –8.726 and statistically significant. Because the ME surprise is normalized such that a one-unit increase corresponds to a 100 basis point upward revision in the expected policy path over the next year, the estimate implies that a 100 basis point hawkish surprise is associated with an 8.7 percent decline in equity prices. More realistically, a 10 basis point surprise leads to approximately a 0.87 percent decline in the S&P 500 during the event window.

Table 1: S&P 500 response to ME

	Dependent variable: SP500
	(1)
ME	-8.726***
	(1.878)
const	0.001
	(0.051)
Observations	256
R^2	0.156
Adjusted R^2	0.153
Residual Std. Error	0.819 (df=254)
F Statistic	21.593*** (df=1; 254)
Note:	*p<0.1; **p<0.05; ***p<0.01

Figures 2 and 3 illustrate the response of nominal Treasury and TIPS yields to a hawkish monetary event surprise of 100 basis points. Nominal yields increase across all maturities, with the largest response occurring at the two-year maturity. The effect declines for longer maturities, suggesting that markets interpret the shock as primarily affecting the near-term path

of policy rates rather than permanently shifting long-run rate expectations. This pattern is consistent with the expectations hypothesis, under which intermediate maturities are most sensitive to revisions in expected short-term rates.

TIPS yields exhibit a similar but declining positive response across maturities, indicating that the shock operates largely through increases in real interest rates. The attenuation of the effect at longer maturities suggests that the tightening is perceived as temporary and does not substantially alter long-run real rate or inflation expectations.

Figure 2: ME Surprise on Treasury Yields

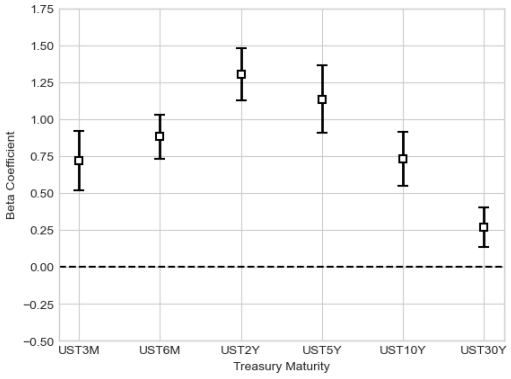
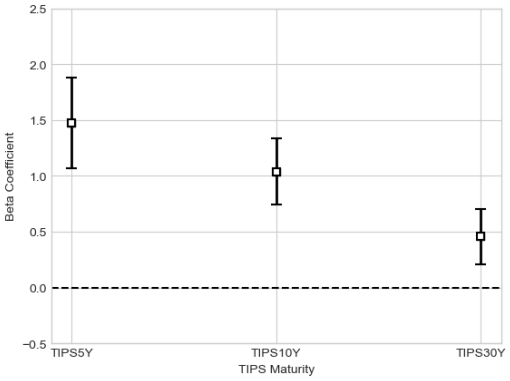


Figure 3: ME Surprise on TIPS Yields



The results above document the average response of the S&P 500, as well as Treasury and TIPS yields, to ME surprises. However, FOMC announcements contain not only policy actions but also forward-looking communication that may influence how markets interpret those actions. Consider two meetings where the Fed raises rates by 25 basis points. If in one meeting the accompanying statement emphasizes that "inflation remains elevated and further increases may be warranted," while in the other it signals that "the pace of increases will slow as conditions evolve," markets may respond very differently despite the identical policy move. In other words, the tone of the statement may reinforce or offset the price impact of the policy action itself, causing asset price responses to vary systematically with the sentiment embedded in the communication. To explore this channel, I incorporate FinBERT-based sentiment scores into the monetary event window data.

Figure 4: FOMC Statement Sentiment Scores

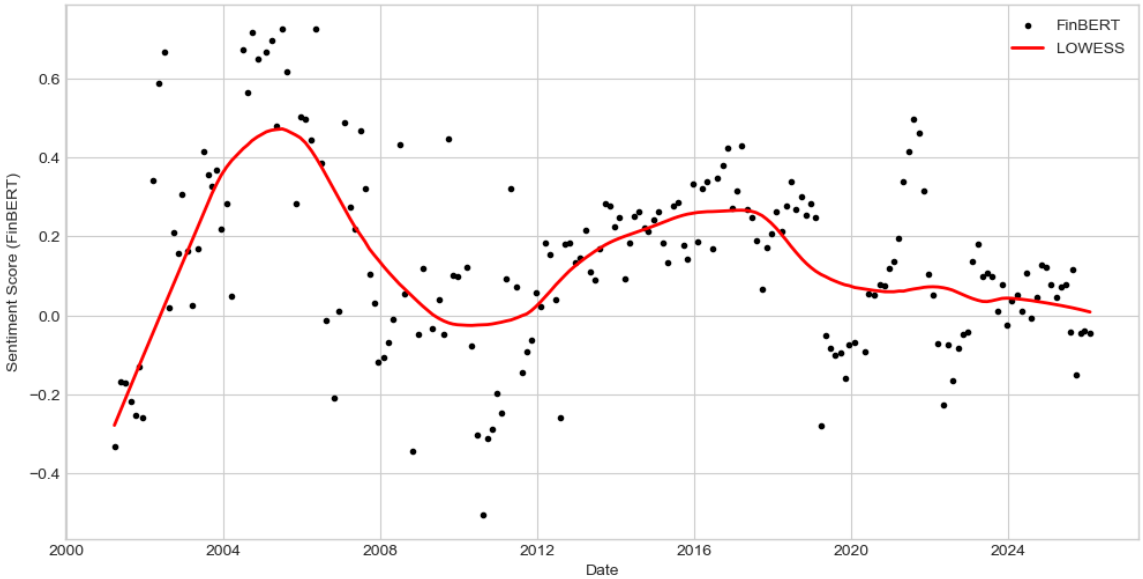


Figure 4 plots the sentiment score for each FOMC statement from 2000 to the most recent, with a LOWESS smoother overlaid to highlight the trend. Several patterns are consistent with the historical record. Sentiment is notably positive in the early 2000s, reflecting the accommodative tone adopted following the dot-com bust and 9/11. It declines sharply through 2007–2011, bottoming out around the financial crisis and its aftermath, when Fed statements were dominated by language around deteriorating conditions and heightened uncertainty. Sentiment recovers gradually through the mid-2010s as the economy expanded and the Fed began normalizing policy, before declining again around 2019–2020 with the COVID shock. The post-2020 period shows scores clustering near zero, consistent with the more neutral, data-de pendent tone the Fed adopted during the tightening cycle. The sentiment series following the broad arc of monetary policy stances over the past two decades lends face validity to the FinBERT-based measure before turning to the regression analysis.

I first estimate a regression of S&P 500 returns on the ME surprise and FinBERT based sentiment scores reported in Table 2. Consistent with the baseline specification, the coefficient on ME remains negative and highly statistically significant. The estimate of –10.229 implies that a 100 basis point hawkish revision in the expected policy path is associated with about a 10.2 percent decline in equity prices within the event window. A more typical 10 basis point surprise corresponds to approximately a 1.02 percent decline in the S&P 500.

Importantly, the coefficient on the sentiment score is positive and statistically significant at the 10 percent level. The estimate of 0.299 suggests that more optimistic communication is associated with higher equity returns, holding constant the mechanical rate surprise. This indicates that forward-looking tone contains incremental information beyond the measured policy shock itself. Together, these results imply that both the unexpected policy path and the qualitative framing of that policy contribute to equity market reactions.

Table 2: S&P 500 response to ME surprise and sentiment

	Dependent variable: SP500
	(1)
ME	-10.229***
	(1.393)
score_FinBERT	0.299*
	(0.181)
const	-0.012
	(0.058)
Observations	190
R^2	0.259
Adjusted R^2	0.251
Residual Std. Error	0.599 (df=187)
F Statistic	28.227*** (df=2; 187)
Note:	*p<0.1; **p<0.05; ***p<0.01

Having shown that both the monetary event surprise and statement sentiment influence equity returns, I next examine whether this pattern extends to fixed-income markets. Specifically, I estimate the same augmented specification using changes in the 10-year Treasury yield as the dependent variable. This allows for a direct comparison of how monetary surprises and communication tone transmit across asset classes.

As reported in Table 3, the coefficient on ME remains positive and highly statistically significant. The estimate of 0.737 implies that a 100 basis point upward revision in the expected policy path is associated with a 73.7 basis point increase in the 10-year yield within the event window. A more typical 10 basis point surprise corresponds to roughly a 7 basis point increase.

In contrast to the equity results, the sentiment coefficient is economically small and statistically insignificant. This suggests that, while equity markets respond to both the policy shock and the tone of communication, long-term Treasury yields primarily react to the quantifiable revision in the expected policy path.

Table 3: UST10Y response to ME surprise and sentiment

	Dependent variable: UST10Y
	(1)
ME	0.737***
	(0.101)
score_FinBERT	-0.003
	(0.011)
const	-0.004
	(0.003)
Observations	190
R^2	0.357
Adjusted R^2	0.350
Residual Std. Error	0.035 (df=187)
F Statistic	27.725*** (df=2; 187)
Note:	*p<0.1; **p<0.05; ***p<0.01

The previous specification examined whether sentiment exerts an independent effect on asset prices. However, communication tone may instead condition the impact of directional monetary policy shocks rather than simply shifting asset prices on its own. A hawkish surprise accompanied by an optimistic statement, signaling that the Fed is tightening into a strong economy, may elicit a larger market response than the same hawkish surprise paired with a pessimistic statement, where markets may interpret tightening as a response to deteriorating conditions and have already priced in some of the shock. In other words, sentiment may not just shift the level of asset prices but also alter the sensitivity of markets to policy signals. To test this channel, I introduce an interaction between the directional policy shock (ED4) and the FinBERT-based sentiment measure, allowing the effect of a hawkish or dovish shock to vary with the tone of the accompanying statement.

Table 4 reports the interaction specification for S&P 500 returns. Consistent with earlier results, the coefficient on the ME surprise remains negative and statistically significant, indicating

that hawkish policy surprises are associated with declines in equity prices. The estimated magnitude implies that a 100 basis point upward revision in the expected policy path corresponds to approximately a 10 percent decline in the S&P 500 within the event window.

The FinBERT-based sentiment score remains positive and statistically significant at the 10 percent level, suggesting that more optimistic communication is associated with higher equity returns, holding constant the policy surprise. This indicates that statement tone independently influences equity valuations outside of directional policy shocks.

The interaction between ED4 and sentiment is statistically insignificant. This suggests that, for equities, communication tone does not significantly alter the marginal effect of directional policy shocks. Instead, tone only exerts a direct effect on equity prices.

Table 4: Interaction of directional policy shocks (ED4) and sentiment on the S&P 500

	Dependent variable: SP500
	(1)
ME	-10.021**
	(4.695)
score_FinBERT	0.300*
	(0.168)
ED4	-0.109
	(2.454)
(ED4 * score_FinBERT)	0.064
	(3.474)
const	-0.014
	(0.062)
Observations	190
R^2	0.259
Adjusted R^2	0.243
Residual Std. Error	0.602 (df=185)
F Statistic	14.291*** (df=4; 185)
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 5 reports the interaction specification for the 10-year Treasury yield. The coefficient on ED4 is positive and highly significant, indicating that hawkish directional shocks increase long-term

yields. The coefficient on the ME surprise also remains significant, capturing the broader informational impact of the combined statement and press conference window beyond the directional component.

Most importantly, in contrast to the S&P 500 results, the interaction between ED4 and sentiment is positive and statistically significant at the 1 percent level. This implies that the impact of a directional policy shock on the 10-year yield depends on the tone of the accompanying communication. Specifically, when the statement communication tone is more optimistic, the yield response to a hawkish directional shock is amplified. Conversely, a more pessimistic tone attenuates the yield response, reflecting the non-linearities and potential market corrections observed in recent policy cycles where pessimistic news may already be reflected in prices.

The standalone sentiment coefficient remains insignificant, suggesting that linguistic tone primarily functions as a modifier that shapes how investors interpret directional policy signals rather than acting as an independent driver of long-term yields.

Table 5: Interaction of directional policy shocks (ED4) and sentiment on UST10Y

	Dependent variable: UST10Y
	(1)
ME	-0.625***
	(0.182)
score_FinBERT	0.002
	(0.010)
ED4	0.649***
	(0.095)
(ED4 * score_FinBERT)	0.396***
	(0.141)
const	0.005
	(0.003)
Observations	190
R^2	0.586
Adjusted R^2	0.577
Residual Std. Error	0.028 (df=185)
F Statistic	51.134*** (df=4; 185)
Note:	*p<0.1; **p<0.05; ***p<0.01

The difference in the interaction results across asset classes is consistent with how equity and fixed-income markets process information differently. Treasury yields are primarily driven by expectations about the future path of short-term interest rates and real activity, making them highly sensitive to the precise signal conveyed by Fed communication. When a hawkish surprise is accompanied by an optimistic statement, bond markets interpret this as a credible and sustained tightening cycle, amplifying the yield response. A pessimistic statement alongside the same hawkish surprise, however, suggests the Fed is tightening into weakness, leading markets to expect the cycle to be short-lived and diminishing the yield reaction. Equity prices, by contrast, aggregate expectations across a much broader set of fundamentals such as earnings growth, risk premia, and the long-run discount rate, and are therefore less sensitive to the specific framing of any single policy signal. For equities, the tone of the statement matters as an independent signal about the economic outlook, shifting valuations directly, but it does not systematically alter how investors translate a given rate shock into a change in stock prices. This distinction is consistent with the view that fixed-income markets are more sensitive to the conditional signal embedded in Fed communication, while equity markets respond to sentiment and surprises as separate inputs.

6. Conclusion

This paper examines whether the linguistic tone of FOMC post-meeting statements contains information relevant to financial markets beyond conventional measures of monetary policy surprises. Combining high-frequency surprise data from the USMPD with FinBERT-based sentiment scores, I find consistent evidence that both channels matter, but in ways that differ systematically across asset classes.

For equity markets, the results point to a dual-channel transmission of FOMC communication. The measured policy surprise captures the mechanical revision in expected rate paths, while FinBERT sentiment captures the qualitative forward-looking framing of that path. A more optimistic statement is associated with higher S&P 500 returns, independent of the direction or magnitude of the policy shock. This suggests that equity investors respond not only to what the Fed does but also to how it characterizes the outlook. Consistent with the literature, a 100 basis point hawkish ME surprise is associated with approximately an 8.7 to 10.2 percent decline in equity prices within the event window, depending on model specification.

For Treasury markets, the picture is more nuanced. Sentiment does not independently shift the 10-year yield, but the interaction between directional policy shocks (ED4) and statement tone is positive and highly significant. Specifically, an optimistic statement amplifies the yield response to hawkish directional surprises, while a pessimistic tone attenuates it. This pattern is consistent with a non-linear market correction mechanism: when the Fed signals pessimism, markets may have already priced in a portion of the tightening, muting the incremental impact of a hawkish directional shock. The interaction model also substantially improves explanatory power, raising the adjusted R^2 from 0.350 to 0.577 for the 10-year yield specification, underscoring the importance of accounting for the joint informational content of policy actions and communication tone.

Taken together, these results contribute to the growing literature on central bank communication by demonstrating that textual sentiment and quantitative policy surprises are complementary rather than redundant inputs for explaining asset price dynamics around FOMC events. The findings carry implications for both policymakers and market participants. For the Federal Reserve, the results suggest that the tone of post-meeting statements is not merely descriptive, but that it can be used as a policy instrument. A hawkish rate decision delivered alongside an optimistic statement will transmit more forcefully to fixed-income markets than the same decision paired with cautious language, meaning the Fed can modulate the financial conditions impact of a given rate action through the framing of its communication. Policymakers who ignore this channel risk sending unintended signals that either amplify tightening beyond what is warranted or cushion it below what is intended. For investors and analysts, the findings suggest that standard surprise measures derived from futures markets capture only part of the informational content of FOMC events. Incorporating sentiment alongside rate surprises improves the explanation of both equity and Treasury yield movements, implying that real-time textual analysis of Fed statements can provide incremental value for predicting asset price reactions within the event window. Future research could extend this framework by exploring non-linear or regime-dependent specifications, incorporating press conference sentiment separately from statement sentiment, or examining cross-asset spillovers beyond U.S. Treasuries and equities.

7. References

Acosta, M., Ajello, A., Bauer, M., Loria, F., & Miranda-Aggripino, S. (2025). Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database. Federal Reserve Bank of San Francisco Working Paper 2025-30. **https://doi.org/10.24148/wp2025-30**

Bernanke, B. S., Kuttner, K. N., (2005). What Explains the Stock Market's Reaction to Federal Reserve Policy? *Journal of Finance*, 60(3), 1221-1257. **https://doi.org/10.1111/j.1540-6261.2005.00760.x**

Gürkaynak, R. S., Sack, B., & Swanson, E. T. (2005). Do Actions Speak Louder than Words? The Response of Asset Prices to Monetary Policy Actions and Statements. *International Journal of Central Banking*, *1*(1), 55-93.

Luca, D. O., & Trebbi, F. (2009). Measuring Central Bank Communication: An Automated Approach with Application to FOMC Statements. NBER Working Paper 15367. **https://doi.org/10.3386/w15367**

Osowska, E., & Wójcik, P. (2023). Predicting the reaction of financial markets to Federal Open Market Committee post-meeting statements. *Digital Finance*, *6*, 145-175. **https://doi.org/10.1007/s42521-023-00096-8**

Romer, C. D., & Romer, D. H. (2004). A New Measure of Monetary Shocks: Derivation and Implications. *The American Economic Review*, *94*(4), 1055-1084. **http://www.jstor.org/stable/3592805**

Yang, Y., UY, M. C., & Huang, A. (2020). FinBERT: A Pretrained Language Model for Financial Communications. *arXiv*. **https://doi.org/10.48550/arXiv.2006.08097**